



Computing Infrastructures
January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

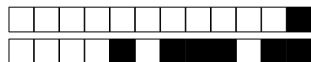
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 2 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A False

☐ B True

Question 3 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A True

☐ B False

Question 4 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A False

☐ B True

Question 5 Containers share the underlying operating system kernel with other containers.

☐ A False

☐ B True

Question 6 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A False

☐ B True

Question 7 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A True

☐ B False

Question 8 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A True

☐ B False

Question 9 A UPS is not necessary for network equipment in a datacenter.

☐ A True

☐ B False

Question 10 SSDs are more energy-efficient than HDDs.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 12000 RPM, negligible controller overhead, and an average seek time of 8 ms. Given that a file of 12 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.82 ($A_{spine} = 0.82$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1300 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8\text{s}$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05\text{ s}$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=10$ concurrent users, each with think time $Z = 120\text{ s}$. What is the maximum response time according to the bound analysis?



Question 15

Determine the maximum throughput and the minimum response time the system can achieve according to the bound analysis.

Question 16

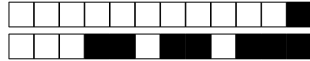
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

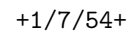
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.





Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

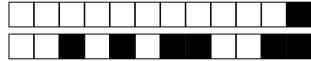
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

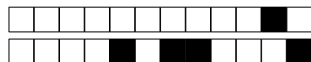
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A False

☐ B True

Question 2 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 3 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A True

☐ B False

Question 4 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A True

☐ B False

Question 5 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A True

☐ B False

Question 6 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A True

☐ B False

Question 7 SSDs are more energy-efficient than HDDs.

☐ A True

☐ B False

Question 8 Containers share the underlying operating system kernel with other containers.

☐ A True

☐ B False

Question 9 A UPS is not necessary for network equipment in a datacenter.

☐ A True

☐ B False

Question 10 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A False

☐ B True



+2/3/48+

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 10000 RPM, negligible controller overhead, and an average seek time of 6 ms. Given that a file of 14 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.82 ($A_{spine} = 0.82$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1200 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8\text{s}$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05\text{ s}$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=10$ concurrent users, each with think time $Z = 120\text{ s}$. What is the maximum response time according to the bound analysis?



Question 15

Determine the maximum throughput and the minimum response time the system can achieve according to the bound analysis.

Question 16

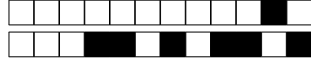
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

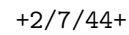
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

[illegible]





Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

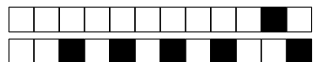
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

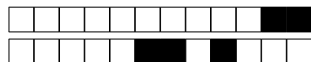
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name: (LAST NAME IN CAPITAL LETTERS)			
First Name: (FIRST NAME IN CAPITAL LETTERS)			

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

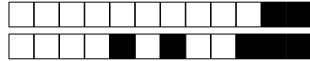
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is **ONLY** the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 2 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A True

☐ B False

Question 3 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A False

☐ B True

Question 4 A UPS is not necessary for network equipment in a datacenter.

☐ A True

☐ B False

Question 5 SSDs are more energy-efficient than HDDs.

☐ A False

☐ B True

Question 6 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A True

☐ B False

Question 7 Containers share the underlying operating system kernel with other containers.

☐ A False

☐ B True

Question 8 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A False

☐ B True

Question 9 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A True

☐ B False

Question 10 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 10000 RPM, negligible controller overhead, and an average seek time of 6 ms. Given that a file of 14 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.82 ($A_{spine} = 0.82$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1300 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8\text{s}$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05\text{ s}$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=10$ concurrent users, each with think time $Z = 120\text{ s}$. What is the maximum response time according to the bound analysis?



Question 15

Determine the maximum throughput and the minimum response time the system can achieve according to the bound analysis.

Question 16

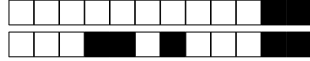
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

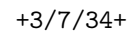
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.

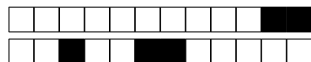


Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

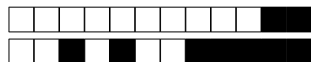
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

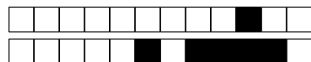
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

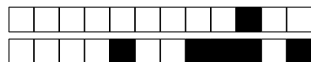
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Containers share the underlying operating system kernel with other containers.

☐ A False

☐ B True

Question 2 SSDs are more energy-efficient than HDDs.

☐ A True

☐ B False

Question 3 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A False

☐ B True

Question 4 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 5 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A True

☐ B False

Question 6 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A True

☐ B False

Question 7 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A False

☐ B True

Question 8 A UPS is not necessary for network equipment in a datacenter.

☐ A True

☐ B False

Question 9 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A False

☐ B True

Question 10 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A True

☐ B False



+4/3/28+

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 12000 RPM, negligible controller overhead, and an average seek time of 8 ms. Given that a file of 12 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.88 ($A_{spine} = 0.88$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1300 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

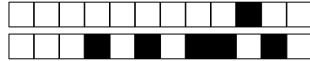
A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8\text{s}$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05\text{ s}$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=8$ concurrent users, each with think time $Z = 120\text{ s}$. What is the maximum response time according to the bound analysis?



Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

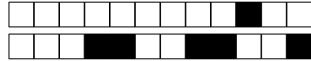
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

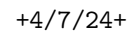
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.





Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

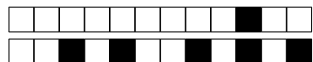
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+4/10/21+

Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

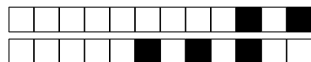
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

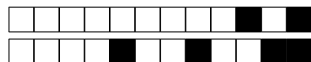
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 2 A UPS is not necessary for network equipment in a datacenter.

☐ A True

☐ B False

Question 3 SSDs are more energy-efficient than HDDs.

☐ A True

☐ B False

Question 4 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A False

☐ B True

Question 5 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A True

☐ B False

Question 6 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A False

☐ B True

Question 7 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A True

☐ B False

Question 8 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A True

☐ B False

Question 9 Containers share the underlying operating system kernel with other containers.

☐ A True

☐ B False

Question 10 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A True

☐ B False



+5/3/18+

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 12000 RPM, negligible controller overhead, and an average seek time of 8 ms. Given that a file of 12 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.88 ($A_{spine} = 0.88$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1300 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8\text{s}$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05\text{ s}$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=10$ concurrent users, each with think time $Z = 120\text{ s}$. What is the maximum response time according to the bound analysis?



Question 15

Determine the maximum throughput and the minimum response time the system can achieve according to the bound analysis.

Question 16

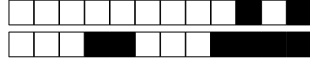
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



+5/6/15+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

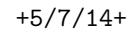
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

[illegible]





Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

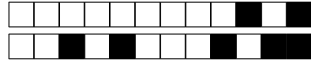
Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+5/10/11+

Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



+6/1/10+

Computing Infrastructures

January 13, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTERS, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping with other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Leaf-spine topologies provide low-latency east-west communication between servers due to the direct connections among the leaf switches.

☐ A False

☐ B True

Question 2 A UPS is not necessary for network equipment in a datacenter.

☐ A False

☐ B True

Question 3 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

☐ A False

☐ B True

Question 4 SSDs are more energy-efficient than HDDs.

☐ A False

☐ B True

Question 5 Hypervisors can only be used to run virtual machines with the same amount of memory as the physical machine on which they are hosted.

☐ A False

☐ B True

Question 6 UPS systems in data centres provide temporary power during grid outages.

☐ A True

☐ B False

Question 7 SSDs have a limited number of read cycles, which can cause them to wear out over time.

☐ A True

☐ B False

Question 8 Edge computing can be used to enable real-time decision-making and improve the responsiveness of applications.

☐ A True

☐ B False

Question 9 Containers share the underlying operating system kernel with other containers.

☐ A True

☐ B False

Question 10 Blade servers can be more expensive and may introduce vendor lock-in due to proprietary enclosures.

☐ A True

☐ B False



+6/3/8+

Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider a hard disk drive (HDD) with the following characteristics: block size of 0.5KB, block transfer time of 0.4 ms, rotation speed of 10000 RPM, negligible controller overhead, and an average seek time of 6 ms. Given that a file of 14 KB must be transferred, and the total I/O time is 31 ms, determine the average access locality that was observed.

Question 12

In a data center network based on a pure leaf-spine architecture without oversubscription, where each switch has 4 ports, what is the maximum number of servers that can be interconnected? If we consider that the leaf switches are ideal ($A_{leaf} = 1$), while the spine switches are each one with an Availability equal to 0.82 ($A_{spine} = 0.82$), what is the availability of the network? Notes: (i) Use at least 4 decimal digits for all the intermediate calculations; (ii) All the other components within the server can be considered as ideal.



Question 13

A vendor sells rack power modules with an MTTF of 1200 days and provides a 2-year warranty. During the warranty period, any failed module is replaced free of charge, at an internal cost of 1.2KEuro each (including materials, logistics, and labor cost). The vendor plans to deliver a total of 400 modules. The company's quality target is that no more than 10% of the modules should fail within the warranty period. i) What is the expected total cost of warranty replacements in KEuro? ii) Is the 10% quality constraint satisfied (YES/NO)?

Question 14

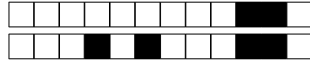
A chatbot platform is modeled as a closed system with two server layers:

- Inference Server (IS): runs the LLM to generate responses.
- Authentication Server (AS): validates user sessions / tokens before requests reach the model.

From the logging system, you extract:

- Service time: $S_{IS} = 0.8s$, Visits: $V_{IS} = 30$ visits per conversation
- Service time: $S_{AS} = 0.05 s$, Visits: $V_{AS} = 120$ visits per conversation

Additionally, the system serves $N=8$ concurrent users, each with think time $Z = 120 s$. What is the maximum response time according to the bound analysis?



+6/5/6+

Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

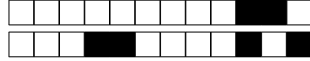
Given the chatbot system described in Question 14, you now have the opportunity to improve its performance by adding exactly one additional server. However, you are allowed to duplicate only one of the existing server layers IS or AS.

The new server will be identical to the existing one of the same type (homogeneous), and incoming requests can be evenly distributed between the two replicas of that server. Assume that:

- Service times of each server type remain unchanged
- Visit ratios are evenly split across replicated servers
- The number of users N and think time Z remain the same

Answer the following:

- a Which server would you choose to duplicate (IS or AS) in order to improve the system bounds? Justify your choice.
- b What is the new minimum system response time (in seconds or minutes) after adding the server?



+6/6/5+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

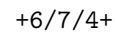
⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

Question 18

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ Compare para- and full virtualization in terms of architecture, operational principles, and performance implications. Describe the advantages and disadvantages of each approach and explain how the different layers of the virtualization stack interact in both models.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



●

⇒ Describe how the requirements of AI/ML applications (in terms of latency, network bandwidth, GPU synchronization, workload variability) directly influence the physical design of the data center (e.g., networking, physical layout, cooling, and power infrastructure).



Computing Infrastructures - January 13, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

Question 12 :

Question 13 :

Question 14 :

Question 15 :

Question 16 :



+6/10/1+

Computing Infrastructures
January 13, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.